

Experiment#07

Objective:

Implementation of circuit diagram of D-Flip Flop and JK Flip Flop. Also write down the difference between both Flip Flop with truth table.

Apparatus:

- IC
- LED
- DMM
- Bread board
- Connecting wires
- NI-Multisim

Theory:

Flip Flop is an electronic device or to be precise a kind of memory component that can hold one bit of data. A flip flop has two states, that is “SET” and “RESET”. Those states are represented with the binary values 0 and 1. The flip flop remains in its current state until it receives a signal that switches it to opposite state. A clock or pulse signal may “trigger” the flip flop to change state.

D-Flip Flop:

D flip flop is an electronic devices that is known as “delay flip flop” or “data flip flop” which is used to store single bit of data. D flip flops are synchronous or asynchronous. The clock signal is required for the synchronous version of D flip flops but not for the asynchronous one. The D flip flop has two inputs, data and clock input which controls the flip flop. When clock input is high, the data is transferred to the output of the flip flop and when the clock input is low, the output of the flip flop is held in its previous state.

JK-Flip Flop:

JK Flip-Flop is basically a gated SR flip-flop which has an additional input that is clock input. It prevents the invalid output that may be obtained when both the inputs are 1. The J.K Flip Flop has two data inputs (J and K) along with clock input and outputs Q and Q'. It can function as a toggle flip flop, where Q toggles when both J and K are high (1). It can also function as a Set-Reset (SR) flip flop when J=1 and K=0, it sets the output Q to 1; when J=0 and K=1, it resets Q to 0. The circuit diagram of a J.K Flip Flop includes J and K inputs, clock input, and outputs Q and Q'.

Difference:

Criteria	D Flip Flop (DFF)	J.K Flip Flop (J.K FF)
Input	Single data input (D)	Dual data inputs (J and K)

Criteria	D Flip Flop (DFF)	J.K Flip Flop (J.K FF)
Functionality	Stores data at input D on clock pulse	Can toggle, set, or reset based on J and K inputs
Number of Outputs	Two (Q and Q')	Two (Q and Q')
Triggering	Triggered by rising edge of clock signal	Triggered by both rising and falling edges of clock
Operation	Follows input D	Depends on J and K inputs

Procedure:

- Set up the D Flip Flop and J.K Flip Flop circuits on a breadboard following the provided circuit diagrams.
- Provide appropriate connections for clock input and data inputs.
- Apply power to the circuits and provide clock pulses.
- Observe the output states of the flip flops based on the input data and clock pulses.

Truth table:

D-Flip Flop:

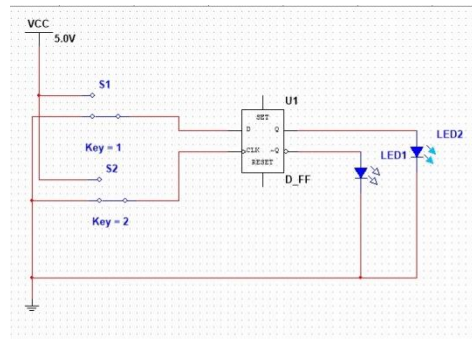
Clock (Clk)	Data (D)	Q (Output)	Q' (Complement Output)

JK-Flip Flop:

Clock (Clk)	J	K	Q (Output)	Q' (Complement Output)

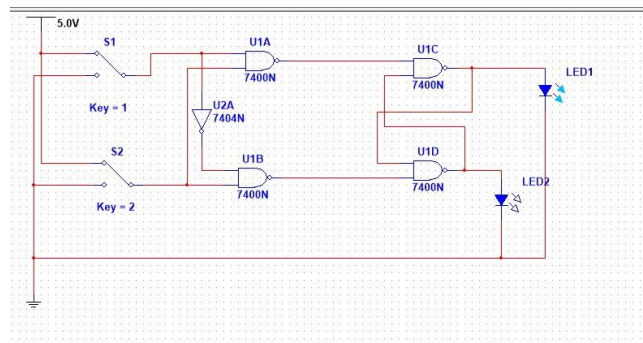
Circuit Diagram:

Symbolic Circuit:



Fig#01

Simple Circuit:



Fig#02

Conclusion:

In this experiment, we know the implementation and understanding of D-Flip Flop and J.K-Flip Flop circuits were achieved. The differentiation between their functionalities and the design of their circuits were successfully demonstrated. The filled truth tables provide a clear understanding of the output states under different input conditions.